

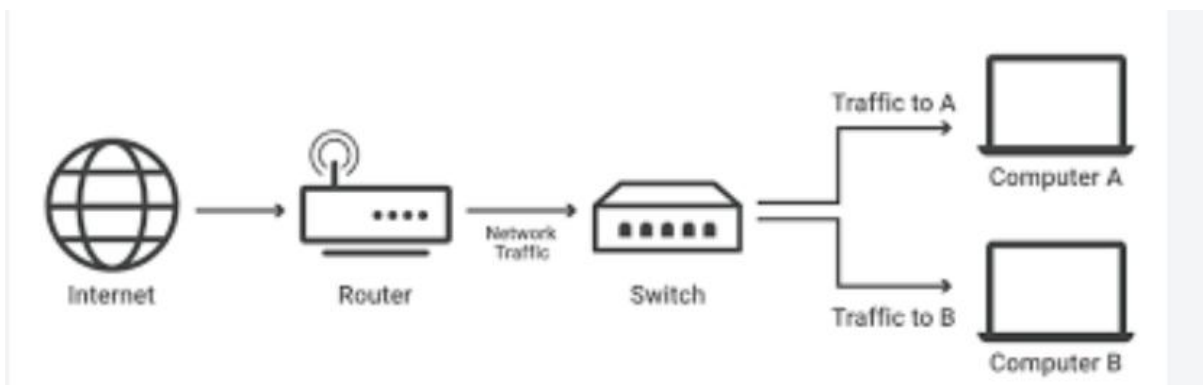
1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

1. Home Wi-Fi Router Layout



- WAN Port (Wide Area Network):
 - Location: Found on the rear panel, typically color-coded in Blue or Yellow and structurally isolated from other ports.
 - Purpose: Connects your local network to an external broadband internet source, such as a fiber terminal or modem.
- LAN Ports (Local Area Network):
 - Location: Positioned in a tight cluster of 4 sequential RJ-45 sockets on the rear panel (often colored Yellow or Black).
 - Purpose: Provides direct high-speed wired access for local network clients like a PC, printer, or smart TV.
- Status LEDs:
 - Location: Integrated across the front or top display panel using small diagnostic light icons.
 - Purpose: Displays critical telemetry at a glance—verifying power status, active internet links, active Wi-Fi radio bands, and showing localized activity bursts.

2. Network Switch Layout



- LAN Ports (Ethernet Ports):

- Location: Form a primary horizontal row or a dense multi-tier matrix grid across the front faceplate.
- Purpose: Serves as high-density interconnection slots where internal local traffic is bridged across your devices at line speed.
- Console Port:
 - Location: Distinctly placed on the outer front edge or back corner, often framed by a light blue color code. On modern enterprise setups, it can be found as an RJ-45 or a Mini-USB slot.
 - Purpose: Provides a dedicated administrative link for network engineers to wire a configuration laptop directly into the device terminal for command-line setup (CLI).
- Status LEDs:
 - Location: Positioned directly above or integrated into the physical corners of every single Ethernet port slot.
 - Purpose: Pinpoints explicit link metrics per wire; showing a steady glow to indicate a successful electrical connection and flickering rapidly to indicate active data frames passing through.

2. List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.
Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

📌 Network Type: A switch connects devices within the same local network (LAN). A router connects entirely different networks together (like your home network to the internet).

📌 Addressing: A switch reads physical MAC addresses to find devices on the local network. A router reads logical IP addresses to locate destinations across different networks.

📌 Traffic Handling: A switch forwards data as frames directly to a specific device using a MAC table. A router forwards data as packets between networks by choosing the most efficient path using a routing table.

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

The Network Router



- **Power Port:** A small, round barrel jack on the rear panel. It connects to the wall adapter to provide electricity.
- **Ethernet Ports:** A row of RJ-45 slots on the rear panel, split by color:
 - **WAN Port (1 port, Blue/Yellow):** Connects to your modem for internet access.
 - **LAN Ports (2+ ports, Yellow/Grey):** Connects your local wired devices (PCs, consoles).
- **LED Indicators:** Light icons on the front panel:
 - **Power LED:** Shows if the device is on or booting.
 - **Internet/WAN LED:** Solid means online; blinking or orange means no internet.
 - **LAN/Wi-Fi LEDs:** Flash rapidly to show active data traffic.

2. The Network Switch



- **Power Port:** Located on the rear or side. (Small DC jack for home units; a standard 3-prong AC socket for lab/rackmount units).
- **Ethernet Ports:** A long, identical row of numbered ports on the front panel. All ports are interchangeable and used to link multiple local devices together.
- **LED Indicators:** Built directly into the corners or right above each individual port:
 - **System Power:** Indicates the switch has power.
 - **Link/Activity LED:** Solid light means a device is plugged in; flashing means data is moving.
 - **Speed LED:** Different colors (Green vs. Amber) indicate the connection speed (e.g., 1 Gbps vs. 100 Mbps).

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

1. Connection to the ISP Line: Router First

- **The Setup:** Plug the incoming ISP internet line directly into the router's WAN port.
- **Justification:** The router is the mandatory gateway. It receives the single internet connection from your ISP, protects the café with a firewall, and assigns internal IP addresses to all devices. A switch cannot do this.

2. Connecting the 8 Gaming PCs: Through a Switch

- **The Setup:** Run one Ethernet cable from a router LAN port to an 8-port (or larger) Gigabit switch. Then, plug all 8 gaming PCs directly into the switch.
- **Justification:**
 - **Port Availability:** Routers usually only have 4 LAN ports; a switch provides enough ports for all 8 PCs.
 - **Performance & Low Ping:** The switch handles all local multiplayer traffic instantly via MAC addresses without overloading the router. This keeps gaming latency (ping) ultra-low while leaving the router free to focus entirely on downloading smooth IPL video streams.